College code : ADR

COLLEGE OF ENGINEERING ADOOR

(Managed by IHRD, Govt. Of Kerala Undertaking)

MANAKKALA P.O, ADOOR, PATHANAMTHITTA , KERALA – 691551

Email : principal@cea.ac.in, Phone : 04734 231995, 8547005100

Our Team

Dr. John George
Mentor (PoC)
Professor and Head
Department of Electrical and Electronics Engineering

Dr. Jayachandran E. S
Mentor
Professor and Head
Department of Electronics and Communication Engineering

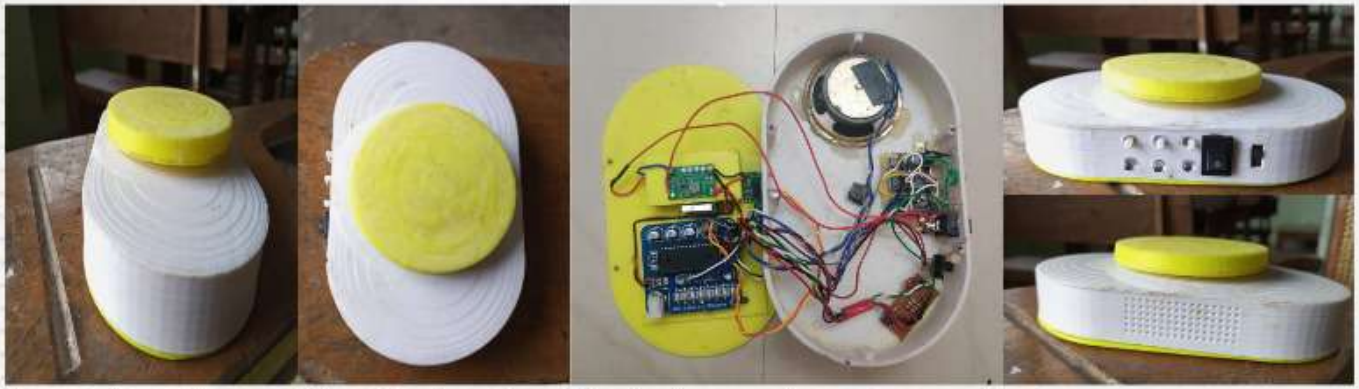
Harijith V Nair

Sajin Sachidanandan

Alif Muhammed

Shebin Alex Oommen

Aneesh A Jose



Short-Term Future Plan: To develop this device to a better one with some more important specifications and implement to the students

Long-Term Future Plan: To develop a much more intelligent device for such children using Machine learning and IoT

Check List of Requirement

Sl. no	Requirements	Yes / No	Remarks
1	RECORD AND PLAYBACK	Yes	
2	AT LEAST 2-MINUTES OF RECORD TIME IN EACH LEVEL	Yes	
3	THREE RECORDING LEVELS.	Yes	
4	LOUD, CLEAR SPEAKER PLAYBACK AND FREE FROM AMBIENT NOISE WHILE RECORDING.	Yes	
5	A CLEAR SNAP-CAP OVER THE TOP OF THE BUTTON TO INSERT PICTURE SYMBOLS.	No	
6	WATER-PROOF, SUITABLE FOR RUGGED USE BY CHILDREN.	Yes	Designed for IPX4
7	ONE SWITCH INPUT TO PROVIDE OPTION FOR ACTIVATION THROUGH ANY OTHER TYPE OF SWITCH, IF REQUIRED BY THE USER.	Yes	
8	ONE TOY/APPLIANCE/EXTENDED OUTPUT SO THAT THE DEVICE WILL SERVE AS A SWITCH TO OPERATE ANY SMART DEVICE OR TO ACTIVATE ANY BATTERY -POWERED TOY.	no	
9	ACTIVATION SURFACE: 10-14 CM (DIAMETER)	Yes	10cm
10	ACTIVATION TYPE: FORCE OF 45-145 GM ON SURFACE	Yes	
11	WEIGHT WITH BATTERY: &LT ;380 GM	No	450gm
12	BATTERY TYPE: 9-VOLT ALKALINE	Yes	
13	UNIVERSAL MOUNTING SOLUTION	Yes	

Cost Analysis

Sl. no	Details	Amount
1	Prototype Cost	6216/-
2	Cost expected for Industrial manufacturing (satisying full requirement) (Large Scale)	2100/-
3	Cost For 50 MOQ	105000/-

Scan the QR Code to get to the video link of the project

part of a waterstreet tourism project.

Tourism hubs) project of the

Engineering colleges develop AAC devices

Prototypes aid children with communication disorders

STAFF REPORTER
THIRUVANANTHAPURAM

Engineering colleges have developed prototypes for augmentative alternative communication (AAC) devices to address the needs of children with communication disorders.

The endeavour is part of the APJ Abdul Kalam Technological University's (KTU) efforts to develop 'Made in Kerala' indigenous AAC devices in collaboration with the National Institute of Speech and Hearing (NISH).

Made in Kerala

Prototypes of augmentative alternative communication devices developed by:
1. College of Engineering, Adoor 2. Muthoot Institute of Technology & Science, Ernakulam, 3. MGM College of Engineering & Technology

Submissions

College of Engineering, Adoor, Vidya Academy of Science and Technology Technical Campus, Federal Institute of Science and Technology, Royal College of Engineering and Technology, Sahrdaya College of Engineering and Technology, MGM College of Engineering and Technology, Muthoot Institute of Science, and Vidya Academy of Science and Technology have submitted the prototypes for evaluation.

KTU Vice Chancellor M.S. Rajasree said the project aligned with the university's vision to utilise the competencies of faculty members, colleges and students to design solutions for the needs of the differently-abled community.

Such efforts are also bound to encourage differently-abled students to pursue engineering courses.

Socially relevant projects

The university has focused on a wider engagement for facilitating solutions for socially relevant projects by utilising the expertise of affiliated colleges and faculty members, she said.

Benefits

Akila Surendran, senior engineer, Centre for Assistive Technology and Innovation, NISH, said early communication intervention with the support of AAC devices can catalyse the development of intentional communication behaviours in children with communication disabilities.

Currently, entry-level AAC devices are imported as there are no companies that manufacture such products in the country.

A single piece could cost around ₹20,000.

Simple, user-friendly

Dr. Surendran said the colleges were given the challenge of developing indigenous version of an entry-level AAC device that was simple, reliable, user-friendly and affordable. The device had to be easily operable by a child with limited motor abilities and rugged enough to withstand rough usage.